ME 453 Hydraulics & Pneumatics

Dr. Yesuenyeagbe A.K. Fiagbe
Department of Mechanical Engineering
KNUST, Kumasi

Maintenance and Safety

General Requirement

- Good equipment properly install
- Properly trained personnel
  - Thorough knowledge of equipment's operations
  - Ability to inspect, Trouble-shoot and remedy problem
- Good maintenance programme
  - Keeping good record
  - Continues inspection
  - System cleanliness
  - Making minor repairs

Tools

- Equipment manuals, blueprints, data sheet, part lists, others
- Adequate testing facility to bench test pumps, controls and actuators
- Adequate tools such as pressure gauge, flow meter, hand tools
- Good inventory of spare parts like seals, bearings, repair kits, extra hydraulic fluid, filter cartridges.

Records

- Equipment record: model number; rating; manufacturer; provided on a card
- Repair card: provide repair history; running cost
- Inspection check list: listing of points or things to be checked and when to check
- A maintenance schedule of inspection
- An inventory control (may be combined with repair history and cost)

Optimum Service Life of Fluid

- Control the Temperature
  - Excessive temperature will oxidize the oil
  - Typical systems Range 43 – 65°C (Some fluid may go upto 120°C)
  - Oil cooler should properly function
  - Air cool radiator should be free of dirt
- Keep System Clean
  - Prevent contamination from diet, water, metal particles
- Keep the Fluid Clean
  - Good storage and handling
  - Inspect fluid filters regularly and change or clean them before they go into bypass mode
- Keep an oil Analysis Programme
  - Regular draining of oil

System Care

- Check fluid level. Add using same oil brand and viscosity grade. DO NOT MIX
- Inspect breather caps, breather filters and fill screen
- Check filter indicators and/or pressure differential gauges
- Visually inspect all system hoses, pipes, pipe connections for leaks.
- Check system temperatures via built in thermometers or otherwise.
- Visually inspect for signs of aeration in the reservoir (use flash light via fill hole).

System Care

- Listen to the pump for signs of cavitation (high-pitched whine or scream)
- Inspect small sample of fluid for colour, signs of contamination and odor.
- Scan electrically controlled servo valves for high temperatures (High valve and solenoid temperatures > 65°C usually indicate the valve is sticking).

SAFETY RULES

- Understand the basic principle and be familiar with components of the system.
- The pressure in the system should never exceed the rated pressure of the lowest rated component.
- Be certain all interfaces to the fluid power system are adequate in strength.
- Never work on or service a system under pressure or while the pump is running (unless bleeding the system)
- Test all circuitry with low pressure before the load is attached.
- Use only the pressure required to achieve the effect.
- Use common sense!!!

Hydraulic Systems Safety

- Always wear eye protection and appropriate clothing.
- Do not remove cylinders until the working units are resting on the floor or ground, on safety stands or on blocks.
- Before disconnecting oil lines, relieve all hydraulic pressure and discharge the accumulator.
- Ensure that all line connections are tight and the lines are not damaged. Escaping oil under pressure is a fire hazard and can penetrate the skin causing serious injury.

Trouble Shooting

- Measurements:
  - Pressure
  - Flow
  - Temperature
- Operating Pressure: good indication if leakage problem occur
  - Less load can be accomplished
  - Actuator may stall under loaded condition

About Pumps

- The hydraulic pump contained within the power pack provides the fluid flow from the power pack. An efficient pump will provide the required flow at system pressure.
- A pump that has excessive internal wear will not provide the required flow at pressure due to internal bypass leakage.
- To keep the power pack operating at it's top efficiency it is necessary to keep the pump in top operating condition.

MOTOR RUNS BUT HYDRAULIC SYSTEM DOES NOT FUNCTION

- Check drive connection between motor and pump, the connecting parts may be broken. Repair or replace if necessary.
- Check for internal failure of pump, deep scoring of pump body, broken pump shaft or gear may be the cause of the problem.
- Repair or replace if necessary.

SLOW OPERATION OF HYDRAULIC SYSTEM

- Check for worn pump, pump with excessive internal wear will not deliver proper system flow at pressure. Inspect pump for wear, repair or replace if necessary.
- Check for too small pump displacement, System should be designed with proper sized pump to match motor size. It may be necessary to increase motor and pump size to increase speed of hydraulic system operation.
- Check for proper line sizes, too small hydraulic lines will restrict pump flow to working cylinders or motors. Flow will be greatly affected by cold weather and small lines. Increase line size if necessary.
- Check for excessive line length, too long lines will affect pump flow due to line flow restriction. Correct if necessary.
- Check for proper relief valve setting, relief valve may be set too close to required system operating pressure. Reset relief valve to correct pressure.

HYDRAULIC SYSTEM RUNS HOT

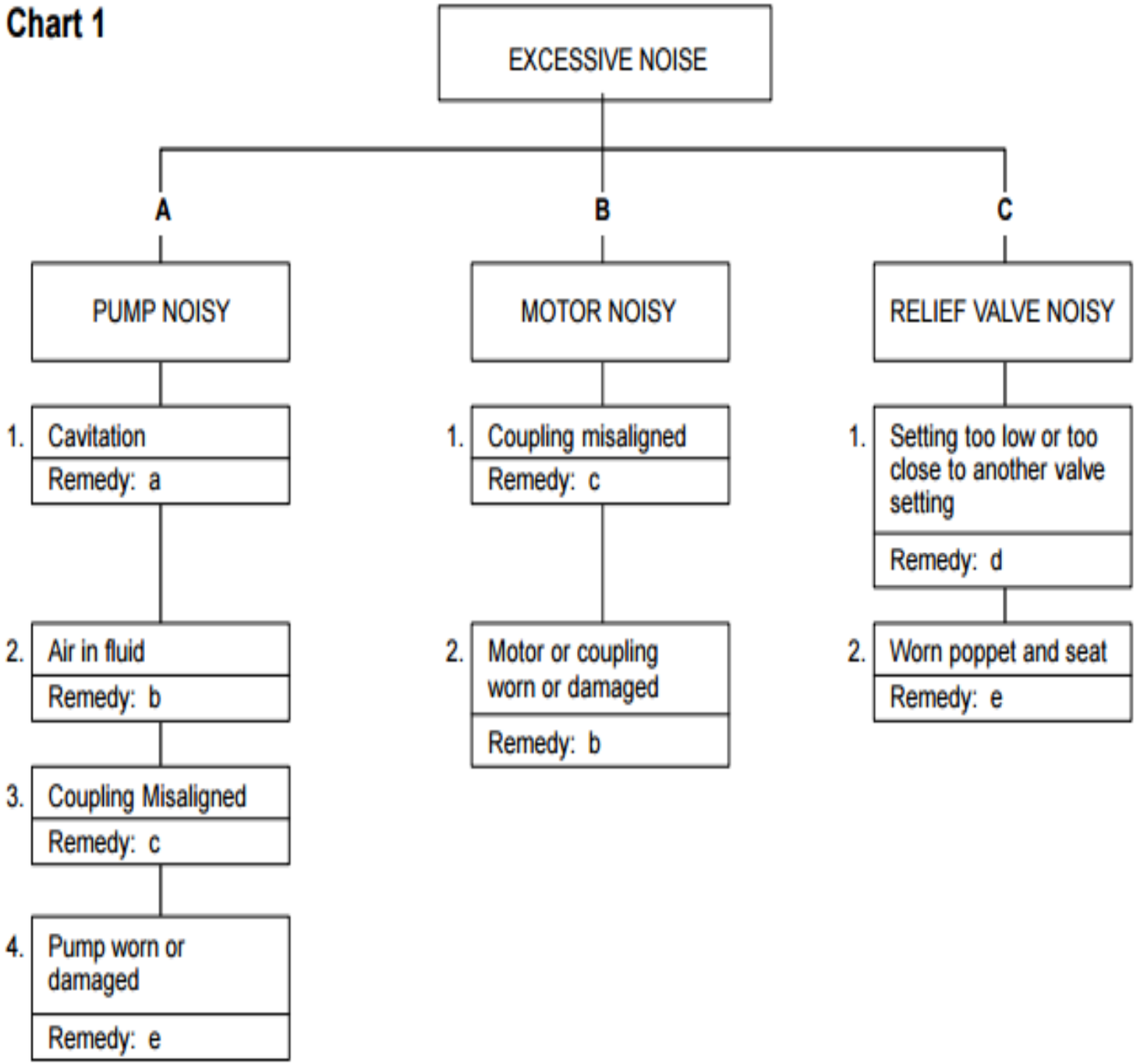
- Check for worn pump – internal wear will cause excessive internal fluid leakage at pressure, resulting in excessive heat generation.
- Inspect pump, repair or replace if necessary.
- Check for correct pump inlet line size as cavitation can cause noise and heat.

SHORT PUMP LIFE

- Check for contaminated oil in system, dirty oil will wear out a pump prematurely.
- To help prevent premature pump failure, change system oil frequently and change system filters on a routine schedule.
- Check oil type, use a quality oil specified for use in hydraulic systems, with the proper viscosity and anti wear properties.
- Check for water in system, water will quickly destroy a pump. Drain system and change to proper oil.

TROUBLESHOOTING GUIDELINES

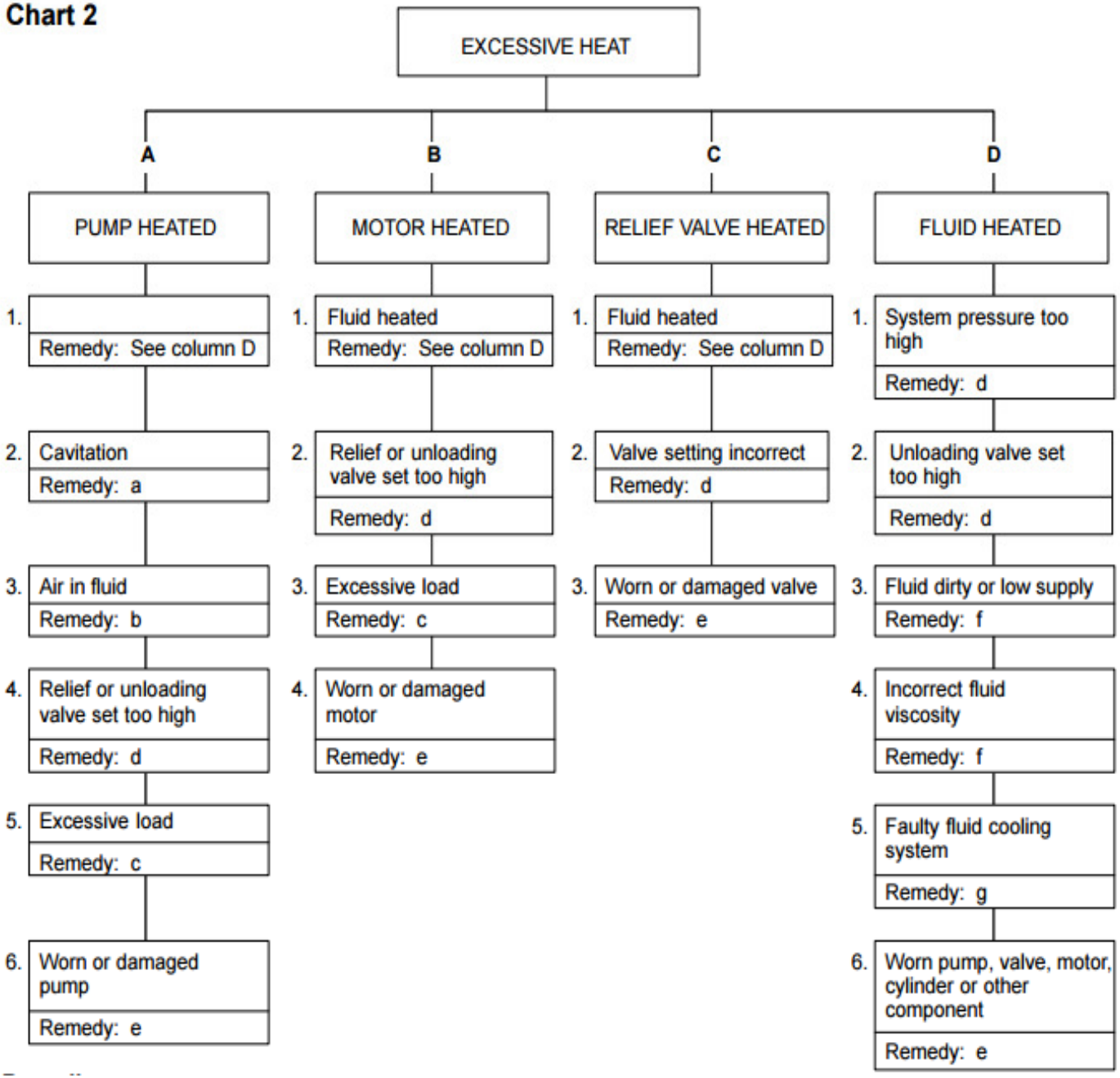
Chart 1


EXCESSIVE NOISE
A
B
C
PUMP NOISY
MOTOR NOISY
RELIEF VALVE NOISY
1. Cavitation
Remedy: a
2. Air in fluid
Remedy: b
3. Coupling Misaligned
Remedy: c
4. Pump worn or damaged
Remedy: e
1. Coupling misaligned
Remedy: c
2. Motor or coupling worn or damaged
Remedy: b
1. Setting too low or too close to another valve setting
Remedy: d
2. Worn poppet and seat
Remedy: e

Remedies (Chart 1)

- a. Any or all of the following: Replace dirty filters; wash strainers in solvent compatible with system fluid; clean clogged inlet line; clean or replace reservoir breather vent; change system fluid; change to proper pump drive motor speed; overhaul or replace supercharge pump; fluid may be too cold.
- b. Any or all of the following: Tighten leaking connections; fill reservoir to proper level (with rare exception all return lines should be below fluid level in reservoir); bleed air from system; replace pump shaft seal (and shaft if worn at seal journal).
- c. Align unit and check condition of seals, bearings and coupling.
- d. Install pressure gauge and adjust to correct pressure.
- e. Overhaul or replace.

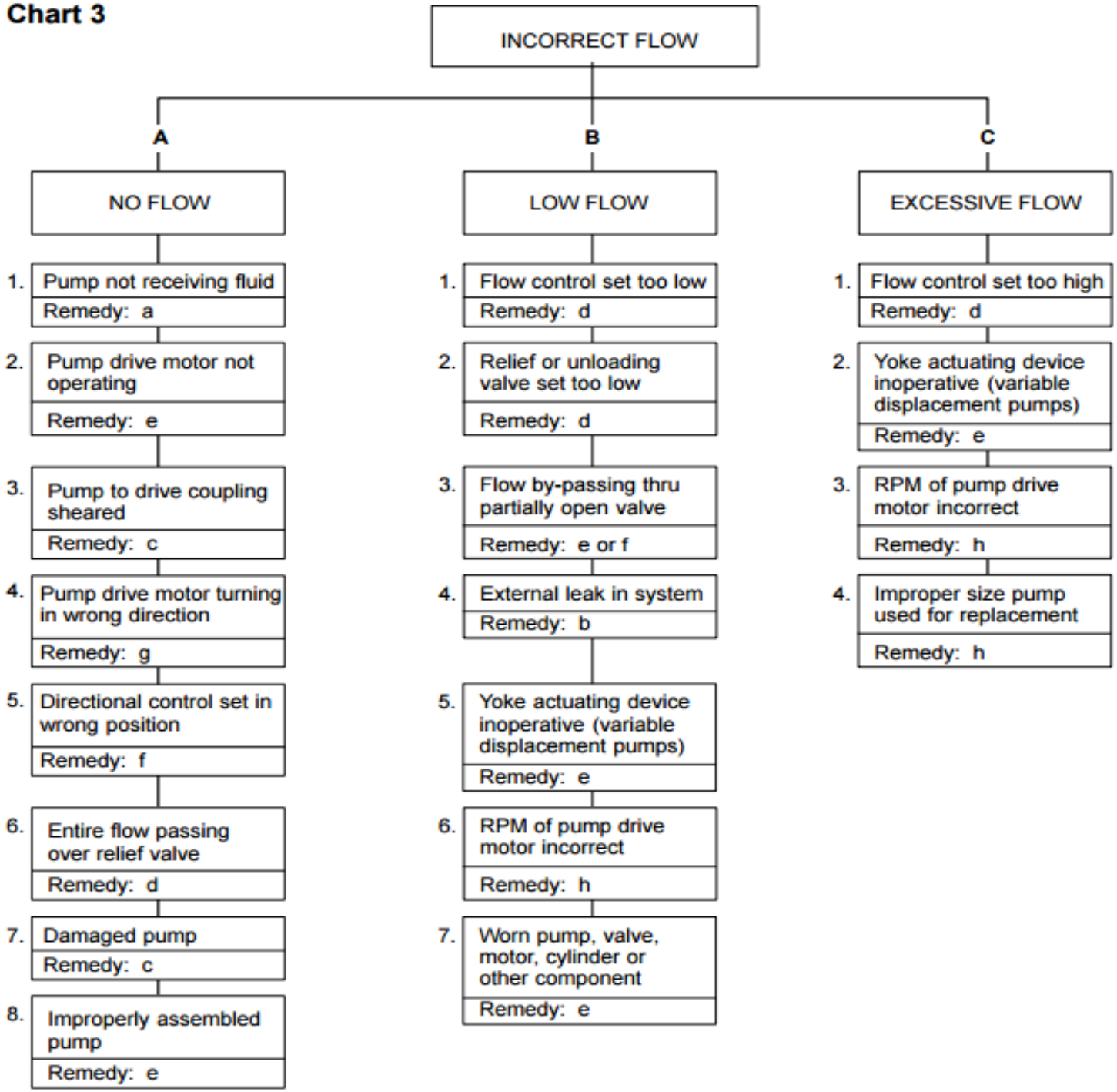
Chart 2


EXCESSIVE HEAT
A
B
C
D
PUMP HEATED
MOTOR HEATED
RELIEF VALVE HEATED
FLUID HEATED
1.
Remedy: See column D
2. Cavitation
Remedy: a
3. Air in fluid
Remedy: b
4. Relief or unloading valve set too high
Remedy: d
5. Excessive load
Remedy: c
6. Worn or damaged pump
Remedy: e
1. Fluid heated
Remedy: See column D
2. Relief or unloading valve set too high
Remedy: d
3. Excessive load
Remedy: c
4. Worn or damaged motor
Remedy: e
1. Fluid heated
Remedy: See column D
2. Valve setting incorrect
Remedy: d
3. Worn or damaged valve
Remedy: e
1. System pressure too high
Remedy: d
2. Unloading valve set too high
Remedy: d
3. Fluid dirty or low supply
Remedy: f
4. Incorrect fluid viscosity
Remedy: f
5. Faulty fluid cooling system
Remedy: g
6. Worn pump, valve, motor, cylinder or other component
Remedy: e

Remedies (Chart 2)

- a. Any or all of the following: Replace dirty filters; clean clogged inlet line; clean or replace reservoir breather vent; change system fluid; change to proper pump drive motor speed; overhaul or replace supercharge pump.
- b. Any or all of the following: Tighten leaking connections; fill reservoir to proper level (with rare exception all return lines should be below fluid level in reservoir); bleed air from system; replace pump shaft seal (and shaft if worn at seal journal).
- c. Align unit and check condition of seals and bearings; locate and correct mechanical binding; check for work load in excess of circuit design.
- d. Install pressure gauge and adjust to correct pressure (keep at least 125 PSI difference between valve settings).
- e. Overhaul or replace.
- f. Change filters and also system fluid if improper viscosity; fill reservoir to proper level.
- g. Clean cooler and/or cooler strainer; replace cooler control valve; repair or replace cooler.

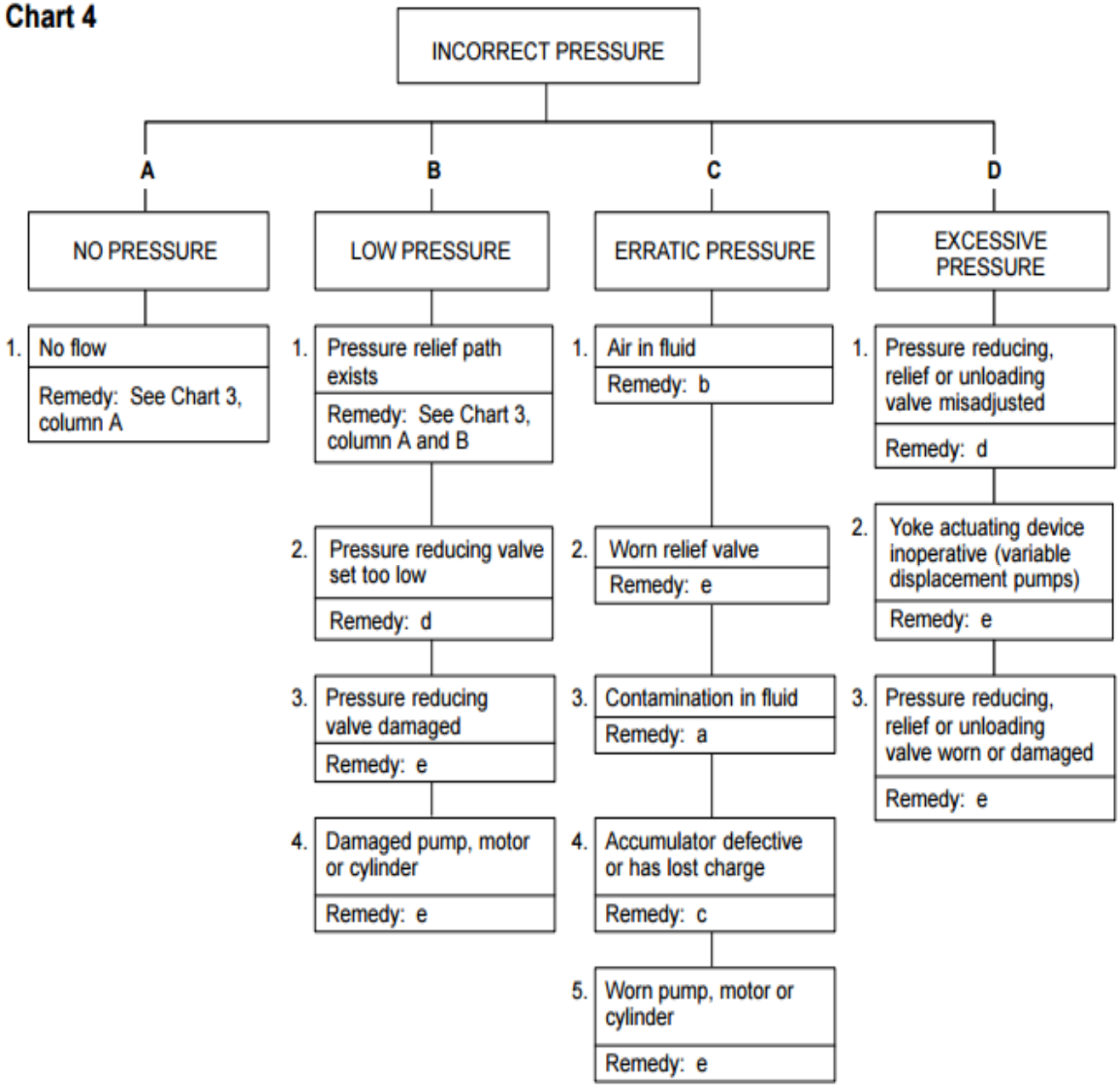
Chart 3


INCORRECT FLOW
A
B
C
NO FLOW
1. Pump not receiving fluid
Remedy: a
2. Pump drive motor not operating
Remedy: e
3. Pump to drive coupling sheared
Remedy: c
4. Pump drive motor turning in wrong direction
Remedy: g
5. Directional control set in wrong position
Remedy: f
6. Entire flow passing over relief valve
Remedy: d
7. Damaged pump
Remedy: c
8. Improperly assembled pump
Remedy: e
LOW FLOW
1. Flow control set too low
Remedy: d
2. Relief or unloading valve set too low
Remedy: d
3. Flow by-passing thru partially open valve
Remedy: e or f
4. External leak in system
Remedy: b
5. Yoke actuating device inoperative (variable displacement pumps)
Remedy: e
6. RPM of pump drive motor incorrect
Remedy: h
7. Worn pump, valve, motor, cylinder or other component
Remedy: e
EXCESSIVE FLOW
1. Flow control set too high
Remedy: d
2. Yoke actuating device inoperative (variable displacement pumps)
Remedy: e
3. RPM of pump drive motor incorrect
Remedy: h
4. Improper size pump used for replacement
Remedy: h

Remedies (Chart 3)

- a. Any or all of the following: Replace dirty filters; clean clogged inlet line; clean or replace reservoir breather vent; fill reservoir to proper level; overhaul or replace supercharge pump.
- b. Tighten leaking connections.
- c. Check for damaged pump or pump drive; replace and align coupling.
- d. Adjust.
- e. Overhaul or replace.
- f. Check position of manually operated controls; check electrical circuit on solenoid operated controls; repair or replace pilot pressure pump.
- g. Reverse rotation.
- h. Replace with correct unit.

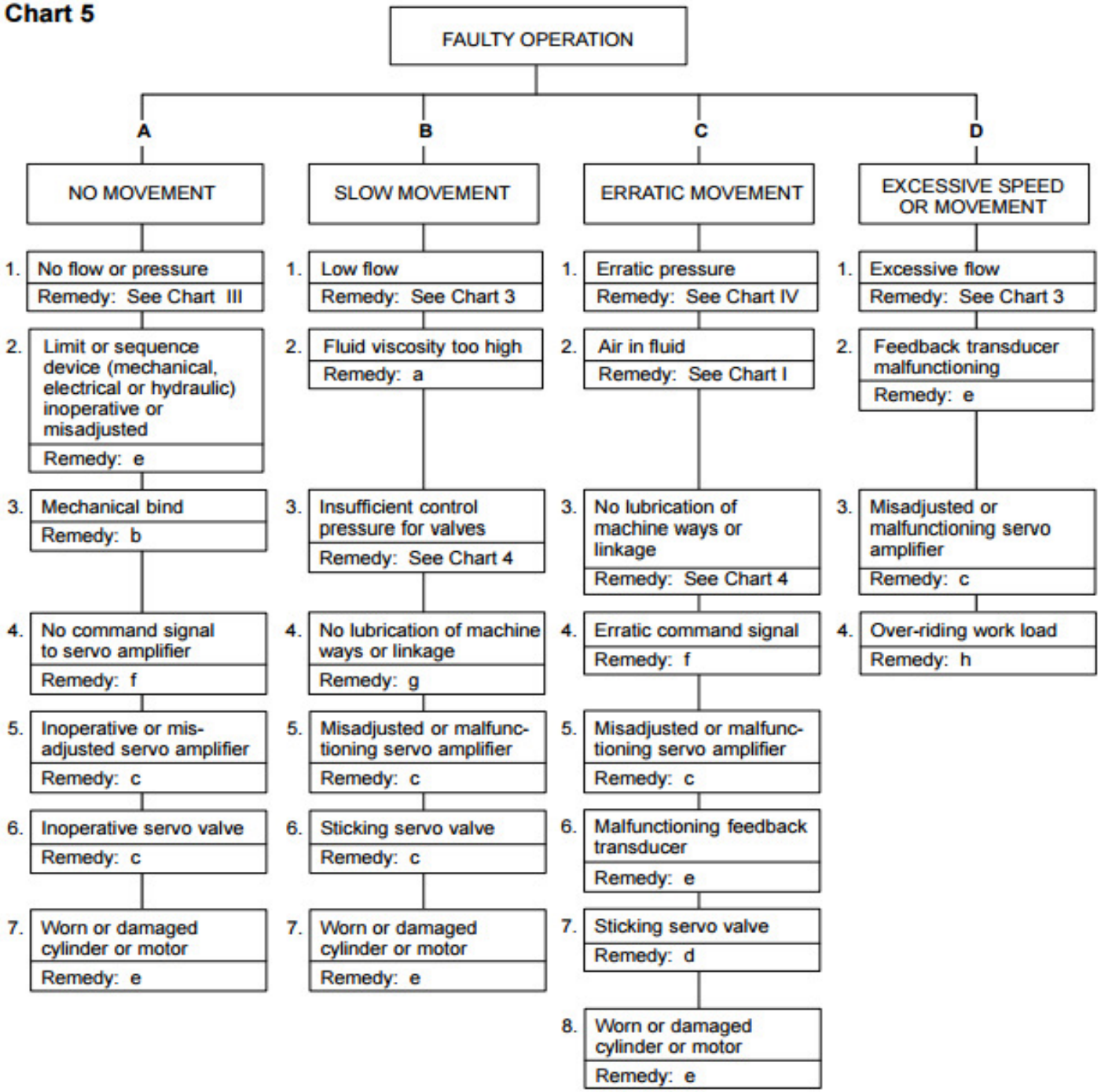
Chart 4


INCORRECT PRESSURE
A
NO PRESSURE
1. No flow
Remedy: See Chart 3, column A
B
LOW PRESSURE
1. Pressure relief path exists
Remedy: See Chart 3, column A and B
2. Pressure reducing valve set too low
Remedy: d
3. Pressure reducing valve damaged
Remedy: e
4. Damaged pump, motor or cylinder
Remedy: e
C
ERRATIC PRESSURE
1. Air in fluid
Remedy: b
2. Worn relief valve
Remedy: e
3. Contamination in fluid
Remedy: a
4. Accumulator defective or has lost charge
Remedy: c
5. Worn pump, motor or cylinder
Remedy: e
D
EXCESSIVE PRESSURE
1. Pressure reducing, relief or unloading valve misadjusted
Remedy: d
2. Yoke actuating device inoperative (variable displacement pumps)
Remedy: e
3. Pressure reducing, relief or unloading valve worn or damaged
Remedy: e

Remedies (Chart 4)

- a. Replace dirty filters and system fluid.
- b. Tighten leaking connections (fill reservoir to proper level and bleed air from system).

Chart 5


FAULTY OPERATION
A
NO MOVEMENT
1. No flow or pressure
Remedy: See Chart III
2. Limit or sequence device (mechanical, electrical or hydraulic) inoperative or misadjusted
Remedy: e
3. Mechanical bind
Remedy: b
4. No command signal to servo amplifier
Remedy: f
5. Inoperative or mis-adjusted servo amplifier
Remedy: c
6. Inoperative servo valve
Remedy: c
7. Worn or damaged cylinder or motor
Remedy: e
B
SLOW MOVEMENT
1. Low flow
Remedy: See Chart 3
2. Fluid viscosity too high
Remedy: a
3. Insufficient control pressure for valves
Remedy: See Chart 4
4. No lubrication of machine ways or linkage
Remedy: g
5. Misadjusted or malfunc-tioning servo amplifier
Remedy: c
6. Sticking servo valve
Remedy: c
7. Worn or damaged cylinder or motor
Remedy: e
C
ERRATIC MOVEMENT
1. Erratic pressure
Remedy: See Chart IV
2. Air in fluid
Remedy: See Chart I
3. No lubrication of machine ways or linkage
Remedy: See Chart 4
4. Erratic command signal
Remedy: f
5. Misadjusted or malfunc-tioning servo amplifier
Remedy: c
6. Malfunctioning feedback transducer
Remedy: e
7. Sticking servo valve
Remedy: d
8. Worn or damaged cylinder or motor
Remedy: e
D
EXCESSIVE SPEED OR MOVEMENT
1. Excessive flow
Remedy: See Chart 3
2. Feedback transducer malfunctioning
Remedy: e
3. Misadjusted or malfunctioning servo amplifier
Remedy: c
4. Over-riding work load
Remedy: h

Remedies (Chart 5)

- a. Fluid may be too cold or should be changed to clean fluid of correct viscosity.
- b. Locate bind and repair.
- c. Adjust, repair or replace.
- d. Clean and adjust or replace; check condition of system fluid and filters.
- e. Overhaul or replace.
- f. Repair command console or interconnecting wires.
- g. Lubricate.
- h. Adjust, repair or replace counterbalance valve.